Experiment 2 Linear and Nonlinear Systems

I. Experiment Objectives

1. Understand the characteristics of limiting, rectifying, and multiplying circuits;
2. Study the behavior of voltage-controlled oscillators (VCOs);
3. Examine the properties of integrators.

II. Lab Requirements

1. Review the definitions of linearity, nonlinearity, additivity, and homogeneity;
2. Review the concept of signal multiplication.

III. Experimental Principles

1. **Limiters and Rectifiers**
   A limiter clips the amplitude of an input signal based on a predefined threshold and outputs the clipped waveform. A rectifier converts an AC input signal into a DC output signal. This experiment investigates the linearity of both devices.
2. **Multipliers**
   A multiplier produces an output signal equal to the multiplication between two input signals. The linearity of the multiplier is examined in this experiment.
3. **Voltage-Controlled Oscillators (VCOs)**
   A VCO is an oscillator whose output frequency is determined by the input control voltage. The operating state and circuit parameters of the oscillator are modulated by the control voltage. This experiment studies the input-output relationship of the VCO.
4. **Integrators**
   An integrator produces an output that is the integral of its input signal. When the input is a square wave, the output can be a triangular or sawtooth wave, enabling waveform conversion. In this experiment, a sawtooth wave is generated using an integrator, and the linearity of the circuit is analyzed.
5. **System Linearity**
   A system is linear if it satisfies both superposition and homogeneity.

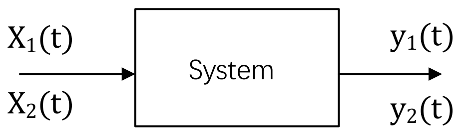

Figure 2.1. System block diagram.

A system is said to satisfy superposition if $x(t) = x_1(t) + x_2(t)$ and $y(t) = y_1(t) + y_2(t)$

A system is said to satisfy homogeneity if $x(t) = ax_i(t)$ and $y(t) = ay_i(t)$, where a is a scalar constant.

IV. Experimental Procedure

1. Characteristics of Limiters and Rectifiers

(1) Limiter

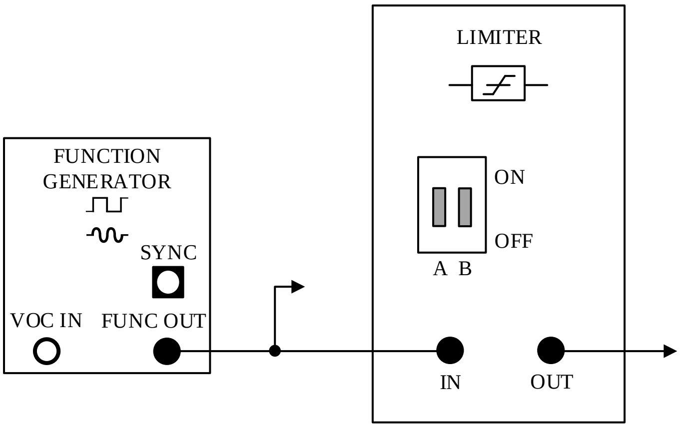

Figure 2.2 Diagram of the limiter system.

Open the SIGEx software, connect the circuit as shown in Figure 2.2 and set the following parameters:

①Function generator: Frequency: 1000 Hz, Amplitude: 1 Vpp, Waveform: Sine

②Limiter: DIP switch set to OFF:OFF

③Oscilloscope: Time base: 4 ms, Trigger: Rising edge on CH0, Trigger level: 0 V

Increase the sine wave amplitude from 1 Vpp to 6 Vpp. Record the input and output voltage readings at each step in Table 2.1.

Table 2.1. Results of the limiter circuit

Input Amplitude （Vpp）	Limiter Output （Vpp）	Rectifier Output （Vpp）
1		
2		
3		
4		
5		
6		

Question 1

Connect **CH0** to **FUNC OUT** and **CH1** to the limiter output. Observe the output waveform.

Is the limiter a linear system? (Yes or No)

(2) Rectifier

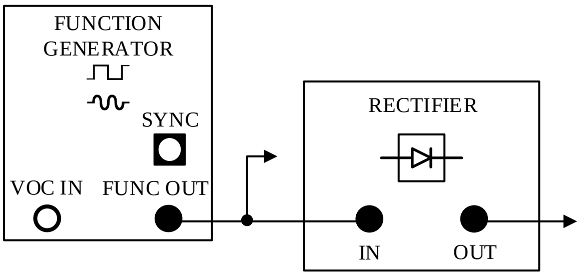

Figure 2.3. Diagram of the rectifier system.

Open the SIGEx software, connect the circuit as shown in Figure 2.3 and set the following parameters:

①Function generator: Frequency: 1000 Hz, Amplitude: 1 Vpp, Waveform: Sine

②Oscilloscope: Time base: 4 ms, Trigger: Rising edge on CH0, Trigger level: 0 V

Increase the sine wave amplitude from 1 Vpp to 6 Vpp. Record the input and output voltage readings at each step in Table 2.1.

Question 2

Connect CH0 to FUNC OUT and CH1 to the rectifier output. Observe the output waveform.

Is the rectifier a linear system? (Yes or No)

(3) Multiplier

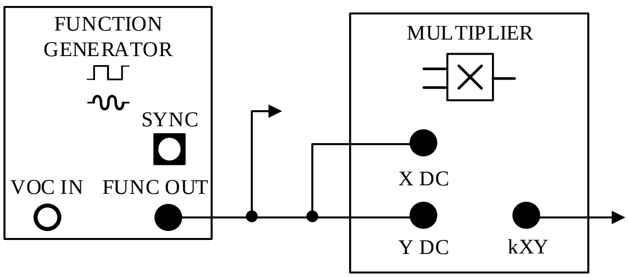

Figure 2.4 Diagram of the multiplier system.

Open the SIGEx software, connect the circuit as shown in Figure 2.4 and set the following parameters:

①Function generator: Frequency: 1000 Hz, Amplitude: 1 Vpp, Waveform: Sine

②Oscilloscope: Time base: 4 ms, Trigger: Rising edge on CH0, Trigger level: 0 V

Increase the sine wave amplitude from 1 Vpp to 10 Vpp. Record the output voltage values at each step in Table 2.2.

Table 2.2 Results of the multiplier circuit.

Input Amplitude（Vpp）	Multiplier Output (Vpp)
1	
2	
3	
4	
5	
6	

Question 3

Connect **CH0** to **FUNC OUT**, **CH1** to the multiplier output. Observe the output signal. Is multiplier a linear system? (Yes or No)

(4) Voltage-Controlled Oscillator (VCO) **(Optional)**

This part investigates the input-output characteristic of a VCO and evaluates its linearity.

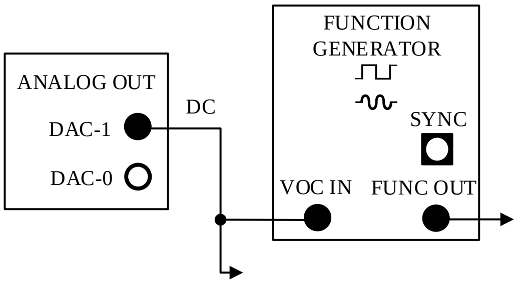

Figure 2.5. Connection diagram of the VCO system.

Launch the SIGEx software, connect the circuit as shown in Fig. 2.5. and set the following parameters:

①Function generator: Frequency: 2000 Hz, Amplitude: 4 Vpp, Waveform: Sine

②Oscilloscope: Time base: 4 ms, Trigger: Rising edge on sine wave output, Trigger level: 0 V

③Modulation type: FM

In Lab 4, adjust the DC input voltage from –3 V to 3 V using the DC knob on the software panel (see Figure 2.7). Ensure the square wave signal switch (Part 4 Signal Select) is **OFF** (green LED off). Measure the output frequency at each DC input voltage. Record the results in Table 2.3.

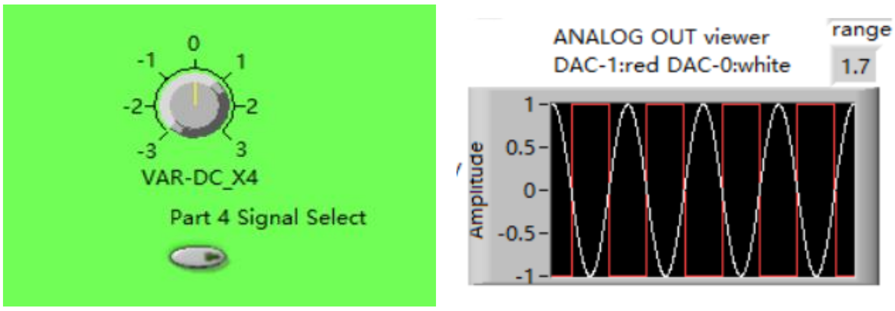

Figure 2.6. Screenshot of the key section in the Lab 4 tab.

Table 2.3. Output characteristics of the VCO system.

DC Input Voltage (V)	VCO Output Frequency (Hz)
-3	
-2	
-1	
0	
1	
2	
3	

Question 4

Connect CH0 to DAC-1 output and CH1 to FUNC OUT. Observe the output signal. Describe the relationship between the input control voltage and the output frequency. Is the VCO response linear over the tested range?

2. Integrator Testing (Optional)

(1) Sawtooth Wave Generator

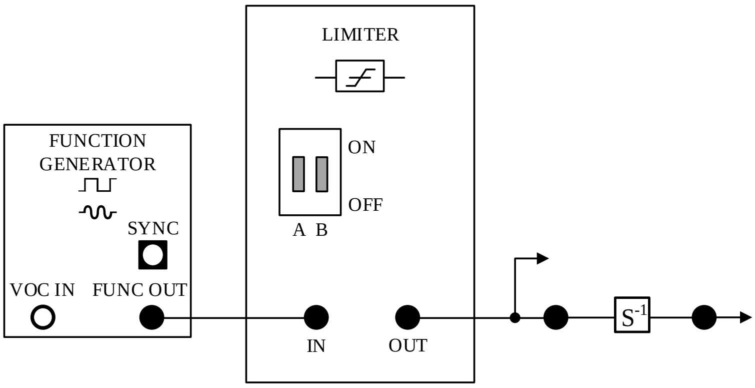

Figure 2.7. Setup diagram of the sawtooth wave generation circuit.

Launch the SIGEx software, connect the circuit as shown in Figure 2.7. and Set the following parameters:

①Function generator: Frequency: 1000 Hz, Amplitude: 4 Vpp, Waveform: Sine

②Oscilloscope: Time base: 2 ms, Trigger: Rising edge, Trigger level: 0 V

③ Modulation type: None

④ Limiter DIP switch: Down ; Down

⑤Integrator rate DIP switch: Up ; Up

Connect CH0 to any input of the continuous-time integrator block (labeled S^{-1}). Connect CH1 to the output of S^{-1}. Observe both waveforms on the oscilloscope. Record the observed signals in your lab notebook.

(2) Linearity Test of the Integrator

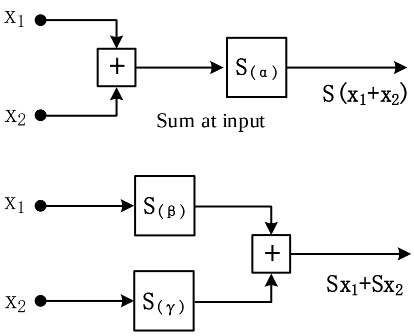

Figure 2.8. Schematic of the integrator linearity test method.

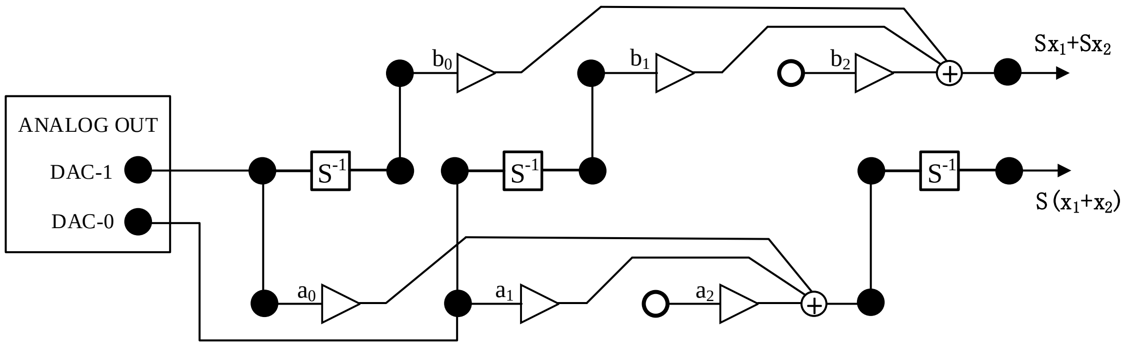

Figure 2.9. Connection diagram for generating both $S(x_1+x_2)$ and Sx_1+Sx_2

Open the NI ELVIS startup interface and select the function generator. Configure the function generator with the following settings: Launch the SIGEx software, Connect the circuit as shown in Figure 2.9. Set the following parameters:

①Integrator rate DIP switch: Up ; Up

②Adder gains (adjusted via software panel):

$a0=a1=b0=b1=1.0$, $a2=b2=0$

③Oscilloscope: Time base: 4 ms, Trigger level: 0 V

In the Lab 4 tab, select the square wave signal as the analog output. Enable the square wave switch (Part 4 Signal Select on the software panel). The green LED should be lit. Refer to Figure 2.6 for the switch location.

Observe the signal before and after integration for $x1+x2$, and record the waveform. Then observe the output of $Sx1+Sx2$, and record the resulting waveform. Plot both waveforms on the same graph. Label each signal clearly.

Question 5(Optional)

Based on the experimental results, does the integrator satisfy the linearity conditions? Adjust the system gains (e.g., set $a_1=b_1$, $a_0=b_0$ and $a_2=b_2=0$) and observe the output. After changing the gains, is the system still linear?

V. Important Notes

1. Prepare all connections before powering on the equipment. After the experiment, disconnect and organize the setup as required. If a power interruption occurs, refer to the recovery procedure in Experiment 1.
2. If any issues arise during the experiment, restart both the hardware and software to resolve them.

VI. Lab Report Requirements

(1) Complete the experiment content independently and record the results honestly.

(2) Answer the experimental questions.